CUSTOMER TRENDS DATA ANALYSIS USING PYTHON, SQL & POWER BI

1. Project Overview

The objective of this project is to analyse customer shopping behavior using Python, MySQL, and Power BI. The analysis helps identify purchasing patterns, customer preferences, and sales trends from retail transaction data. Insights generated from the project support better business decisions related to marketing, product strategy, and customer retention. The project follows the complete data analytics life cycle, including data cleaning, SQL analysis, visualization, and reporting. The final outcome is an interactive dashboard with actionable business insights

2. Dataset Overview

The dataset contains customer shopping behavior collected from a retail business.
Each row represents the latest purchase made by one customer.
The dataset contains information regarding customer demographics, product purchases, payment methods, subscriptions, discounts, shipping preferences, and customer reviews.

Column	Description
customer_id	Unique customer identifier
age	Customer age
gender	Male/Female
item_purchased	Purchased product
category	Product category
purchase_amount	Purchase value
review_rating	Customer rating
payment_method	Payment mode
shipping_type	Shipping method
discount_applied	Whether discount was applied
subscription_status	Subscription status
season	Purchase season
frequency_of_purchases	Purchase frequency

```
df.head()
```

tomer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

3. Data Cleaning & Data Manipulation (Python)

The raw dataset contained inconsistencies, formatting issues, and missing values.
Data preprocessing was performed before analysis.

Step 1: Loading Dataset

The dataset was imported into Python using the Pandas library.

Step 2: Dataset Inspection

The structure of the dataset was examined using:

- df.info()

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- df.describe()

```
df.describe()
# summary statistics of the numerical columns only
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- df.isnull().sum()

  These functions helped identify:
- Data types
- Missing values
- Numerical statistics
- Dataset dimensions

Step 3: Missing Value Treatment

The Review Rating column contained missing values.

Instead of replacing them using the global average, missing ratings were replaced with the median review rating within each product category.

This preserves category-specific customer behavior and reduces bia

```
# To replace the null values => Median over mean because
# Mean     => get affected by outliers that may skew the distribution of the data
# Median   => robust to outliers , continue to give reliable results even when unusual values are present.
```

```
# we'll impute the missing valuees using median review rating of each category
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x : x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

Step 4: Standardizing Column Names

All column names were converted into snake_case.

Benefits:

- SQL compatible
- Easier coding
- Consistent naming convention

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ' , '_' )
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```
df = df.rename(columns={'purchase_amount_(usd)' :'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

Step 5: Feature Engineering

Age Group

Customers were divided into meaningful age groups.

Example:

- Teen
- Young Adult
- Adult
- Senior

This enables customer segmentation.

```
# Feature Engineering
# Create new column age_group
labels = ['Young Adult' , 'Adult' , 'Middle-aged' , 'Senior' ]
df['age_group'] = pd.qcut(df['age'] , q=4 , labels = labels )
```

```
df[['age' ,'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Purchase Frequency

Text values such as

- Weekly
- Monthly
- Quarterly

were converted into numerical values representing purchase frequency in days.

This makes quantitative analysis possible.

```
frequency_mapping  = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Monthly' : 30,
    'Quaterly' : 90,
    'Bi-Weekly' : 14,
    'Annually' : 365,
    'Every 3 months' : 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days' , 'frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14.0	Fortnightly
1	14.0	Fortnightly
2	7.0	Weekly
3	7.0	Weekly
4	365.0	Annually
5	7.0	Weekly
6	NaN	Quarterly
7	7.0	Weekly
8	365.0	Annually

Step 6: Removing Redundant Columns

The column promo_code_used

contained exactly the same information as

discount_applied

Therefore it was removed to eliminate redundancy.

Step 7: Final Clean Dataset

The cleaned dataset now contains:

- No missing review ratings
- Standardized column names
- New engineered features
- Consistent data types
- Reduced redundancy

4. Database Integration & SQL Analysis

After cleaning and preprocessing the dataset in Python, the final dataset was connected to **MySQL** using SQLAlchemy. The cleaned data was imported into a MySQL database, where SQL queries were executed to analyze customer shopping behavior and generate business insights. The analysis included revenue by gender, identifying high-spending customers, customer segmentation, and ranking top-selling products using SQL window functions. MySQL enabled efficient storage, retrieval, and analysis of the data, supporting data-driven business decision-making.

SQL Business Analysis

Several SQL queries were executed to answer business questions.

Revenue by Gender

Identify whether male or female customers contribute more revenue.

gender	revenue
Male	157890
Female	75191

High Spending Customers with Discount

Identify customers whose spending remains above average despite receiving discounts.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79

Customer Segmentation

Customers were classified as:

- New Customers
- Returning Customers
- Loyal Customers

using SQL conditions based on purchase history.

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

Product Ranking using Window Functions

Products were ranked inside each category using

ROW_NUMBER()

This helps identify top-selling products.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Other SQL Analysis

Additional SQL queries included:

- Average purchase amount by category
- Revenue by season
- Most preferred payment methods
- Discount effectiveness
- Subscription analysis
- Shipping performance
- Average customer ratings

5. Power BI Dashboard

- A professional interactive dashboard was developed using Microsoft Power BI.
- The dashboard enables management to analyze customer purchasing behavior dynamically.

KPI Cards

The dashboard includes:

- Total Customers
- Total Revenue
- Average Purchase Amount
- Average Review Rating

Visualizations

The dashboard contains:

Donut Chart

Subscription Status Distribution

Bar Chart

Revenue by Product Category

Column Chart

Revenue by Age Group

Pie Chart

Shipping Preferences

Dashboard Interactivity

Interactive slicers were added for:

- Subscription Status
- Gender
- Category
- Shipping type

Users can dynamically filter the dashboard to explore different customer segments.

6. Key Business Insights

The analysis revealed several valuable business insights.

Insight 1

Customers choosing **Express Shipping** have a higher average purchase amount compared to customers using standard shipping.

Insight 2

Young Adults contribute the highest revenue among all age groups.

Insight 3

Several **Loyal Customers** are still not subscribed.

This presents an opportunity to improve customer retention through targeted subscription campaigns.

Insight 4

Certain product categories consistently outperform others in revenue generation.

These categories should receive greater marketing focus.

Insight 5

Discounts improve purchase frequency but do not always increase overall revenue.

Discount strategies should therefore be optimized carefully.

Insight 6

Higher customer review ratings are generally associated with higher purchase values, indicating customer satisfaction positively influences spending behavior.

9. Conclusion

This project demonstrates the complete end-to-end Data Analytics workflow, beginning with raw customer data and ending with actionable business insights through an interactive dashboard. Python was used for data preprocessing and feature engineering, PostgreSQL enabled advanced SQL-based business analysis, and Power BI provided an intuitive platform for visualizing customer trends. The findings can help retail organizations improve marketing strategies, enhance customer satisfaction, optimize product offerings, and increase overall business profitability.